

Herald



Herald — Status

Field	Value
Revision	17
Created	2026-05-20
Last modified	2026-05-29
Status	active

r17 catches up a week of release-cycle work that the doc had not tracked.

HEAD 3048b48; constitution at 3244bf4 (just gained §11.4.100 Video color + visual-quality fidelity mandate; Herald non-applicable-but-cite — no video-playback surface). Tags shipped since r16: v0.4.0 (Wave 6 closed-loop), v0.5.0 (Wave 6.5 lifecycle), v0.6.0 (Wave 8 Track B MTProto §11.4.98 full-automation). v0.6.0 covers MTProto user-account harness — 3 MTProto-driven live tests (HRD-140/141/142 Subscribe / Wave-6 closed-loop / Wave-6.5 lifecycle) all PASSING LIVE; gotd/td v0.144.0 vendored as submodules/gotd-td per §11.4.74. 3 legacy NON-COMPLIANT tests (TestSubscribe_LiveBotAPI, tests/test_wave6_live_loop.sh, lifecycle --manual) graduated to §11.4.90 Obsolete (HRD-143/144/145) with triple-check evidence. Closed waves since r16: Wave 6 (HRD-100), Wave 6.5 (HRD-101 + 15-scenario lifecycle), Wave 7 multi-channel framework (HRD-110..121 — Slack adapter live), Tgram-completeness (HRD-133..138, 6-HRD wave), Wave 8 Track B (HRD-139..142). v1.0.0 readiness workstream: Batch A (HRD-018/026/027 —

Status
summary

commons_constitution audit write-through + admin REST mode-flip), Batch B (HRD-019..025 — 7 flavor §42 constitution bindings, in_progress until §43 live-op surface lands), Batch C (HRD-029..056 — §43 command catalogue closed in 9 sub-batches per Tasks #195-#204), Batch D (HRD-085..089 — commons_infra pgxTaskRepository), GAP-3 §11.4.85 stress+chaos suite (HRD-122..130 + HRD-132 Runner idempotency-window fix). Inheritance gate **15 PASS / 0 FAIL**. scripts/e2e_bluff_hunt.sh run #1 today reported **148 PASS / 1 FAIL** (E132-anchor) — the 1 FAIL caught a real anti-bluff issue (canonical status=PASS literal missing from fanout/assertion.txt) which was atomically fixed in this session's 3048b48 (test source + committed evidence); bluff-hunt #2 running to confirm 149/0. Workspace at 16 Herald modules + 17 vendored submodules (9 Helix-stack + containers + Concurrency/Models/TOON/gotd-td/http3/slack-go/telebot). HRD-131 §11.4.93 SQLite workable-items SSoT migration Phase 1 remains OPEN-deferred per operator 2026-05-27 (constitution-repo migration binary unimplemented). Prior r16 captures Wave 6 closed-loop runtime. New Cobra subcommand pherald listen runs the closed-loop

Field	Value
	runtime: Telegram getUpdates long-poll (OnText/OnPhoto/OnDocument/OnVoice) + bot self-filter (bot.Me.Username anti-echo guard) + sha256 content-addressed attachment download into ~/ .herald/inbox/ <sha256> . <ext> + Claude Code dispatch with Opus pinned in argv (claude --model claude-opus-4-7 --resume <UUID> --print <envelope>) + envelope pre-text (verbatim operator wording "We have received new message from our communication channel") preceding the <<<HERALD-DISPATCH-v1>>> structured block + <<<HERALD-REPLY>>> parser with action routing + tgram.Adapter.SendReply wiring reply_to_message_id.T10a --qa-out-dir JSONL journaling. 12 commits T1=ad87d7f..T12=96c7c6b. 8 new e2e invariants E63-E70 (all currently SKIP-with-documented-reason — convert to PASS at T10b live evidence). Wave 6 mutation gate tests/test_wave6_mutation_meta.sh (3 paired hermetic). Pre-existing e2e FAILs unrelated to Wave 6 (Postgres SASL on :24100 surfaced in E7-E12 + E14-E16 + E37-E42) tracked separately — NOT Wave 6 regressions. Workspace at 16 modules (Wave 5 added qaherald; Wave 6 added NO new module — inbound is pherald/internal/inbound/). Prior r15 captured Wave 4b TOON (tag v0.3.0). Prior r14 captured Wave 4a HTTP/3+Brotli+Alt-Svc+TLS (tag v0.2.0). r13 closed HRD-016 (Wave 3b pherald Runner live). r12 closed HRD-011 (Telegram live, message_id=5). Every /v1/* business route on every serving flavor now honors Accept: application/toon → real TOON-encoded body (via digital.vasic.toon+toon-format/toon-go upstream wire-up replacing the prior sentinel-error scaffold from the §107 PASS-bluff revert). JSON remains the honest fallback. POST /v1/events accepts both TOON and JSON request bodies (TOON transcoded to JSON before the Runner's EventParser; Runner stays codec-agnostic). HERALD_DEFAULT_RESPONSE_CODEC=json operator opt-out. TOONMiddleware auto-wired into cli.RunServe — every flavor binary (pherald/cherald/sherald/bherald/rherald/iherald/scherald) inherits for free. Real compression evidence: sherald/v1/safety_state TOON=215B vs JSON=226B (~5% on a small payload; larger tabular payloads see 30-60% per upstream measurements). 7 new e2e invariants E56-E62 (E62 SKIP-with-reason — encoder-failure fallback unit-tested in commons/cli/toon_test.go). New tests/test_wave4b_mutation_meta.sh (5/5 PASS — M1 strip TOONMiddleware → E57 FAIL; M2 swap toon.Marshal → json.Marshal (the EXACT 2026-05-17 PASS-bluff signature) → E60 FAIL — gate proves wire-byte-0 anti-bluff catches the recurrence; M3 no-op transcodeRequestBody → pherald unit test FAIL; working-tree quiescence guard (lesson from 72e81ab — script greps own working tree for residual mutation markers BEFORE declaring PASS); post-flight green). Workspace module count unchanged at 15 (no new module). Tag v0.3.0. Prior r14 captured Wave 4a (HTTP/3 + Brotli + Alt-Svc + TLS 1.3; tag v0.2.0). r13 closed HRD-016 (Wave 3b pherald Runner live). r12 closed HRD-011 (Telegram live). r10-r11: Wave 3a substrate. HRD-008, HRD-015, HRD-019..HRD-025 (in_progress — §42 flavor bindings, live-op surface deferred to Batch C), HRD-029..HRD-056 (many migrating to Fixed in v1.0.0 Batch C closures — see Issues.md r36 for canonical state), HRD-081, HRD-085..HRD-090 (in_progress — Batch D live evidence), HRD-115..HRD-121 (Wave 7 T6-T12 — Slack adapter + multi-channel framework, code-complete per Tasks #206-#212, awaiting Issues→Fixed migration audit), HRD-131 (OPEN-deferred per operator). 3 Obsolete-graduated this session: HRD-143/144/145 (legacy NON-COMPLIANT tests).
Issues	

Field	Value
Issues summary	see <code>Issues.md</code> r36 — atomic Obsolete graduation of 3 legacy tests landed alongside Wave 8 Track B closure. ~48 in_progress/open rows remain; many of HRD-029..HRD-056 are now closed per Tasks #195-#204 and need atomic Issues→Fixed migration sweep per §11.4.19 (next-session pickup).
Fixed	HRD-001..HRD-007, HRD-009, HRD-009b, HRD-010, HRD-011, HRD-012, HRD-013, HRD-014, HRD-016, HRD-017, HRD-018, HRD-026, HRD-027, HRD-028, HRD-080, HRD-092..HRD-101, HRD-110..HRD-114, HRD-122..HRD-130, HRD-132, HRD-133..HRD-138, HRD-139..HRD-142 — see <code>Fixed.md</code> r17 for canonical list.
Fixed summary	see <code>Fixed.md</code>
Continuation	QA workstream next: Constitutional addition for <code>docs/qa/</code> evidence mandate + qaherald Go binary driving full pherald  Telegram round-trip automation with attached materials preserved under <code>docs/qa/<run-id>/</code> ; final close-out is tag <code>v1.0.0-dev-0.0.1</code> (or canonical-named) release. Carry-over Wave 3c items (HRD-024 iherald paging, HRD-033 destructive-guard body, HRD-018..027 flavor bindings, the rest of the §43 command catalogue) remain queued. Each milestone followed by multi-mirror push + full anti-bluff verification.

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Project phase

Implementation-r1, Foundation complete + Wave 2 flavor scaffolds landed. Spec V3 r8 is active, the Go module foundation compiles, all three Foundation milestones (M1/M2/M3) plus the Wave 2 flavor-scaffold workstream have landed, and the §107 end-user-usability covenant is now explicit in Herald's three root docs + gate (added 2026-05-20). The remaining cycle is wiring live channel integrations (HRD-011, HRD-016, HRD-024, HRD-028, HRD-098), validating the §26.5 quickstart end-to-end on operator hardware, and implementing the 28 §43 command bodies still stubbed.

Specification

- **Active spec:** `docs/specs/mvp/specification.V3.md` (Revision 8, ~4600 lines after r8 Wave 2 capture).
- **Archived specs:** V1 + V2 in `docs/specs/mvp/archive/` (frozen).
- **Recent sections:** §37 Tracker-doc change events; §38 Workable-item announcement contract; §39 Message presentation + Herald Canonical Template; §40 Documentation + 15 named test challenges; §41 REST API surface (Gin Gonic); §42 Constitution-binding integration; §43 Constitution-derived flavor commands (27 entries → HRD-029..HRD-056) + §43.2 catalogue-check addendum (r8); §44 Foundation implementation contract + §44.M Wave 2 milestone capture (r8); §6.3 Branding extension with per-flavor fields (r8); §18.0 Wave 2 flavor-binary catalogue (r8).

- **Spec-change rule:** §23 + HERALD_CONSTITUTION §106 + I7 gate — green throughout.
- **End-user-usability covenant:** HERALD_CONSTITUTION §107 + I8 gate (added 2026-05-20) — green; paired §1.1 mutation meta-test in `tests/test_i8_usability_meta.sh` (5/5 PASS).

Implementation

Module	Status	Notes
<code>commons</code> (L0)	✓ landed	§11.0 types + Clock + UUIDv7 + DefaultBranding; tests pass.
<code>commons_prefix</code>	✓ landed	§8.2 algorithm + 5 test functions; tests pass. 14 files, ~2.9k LOC — Evaluator + Registry + 12 emit helpers + BundleHash + ModelLadder + ConstitutionStore + MemoryBus + Runner + CloudEvents v1.0 adapter; all green under <code>go test -race</code> .
<code>commons_constitution</code> (M1)	✓ landed	
<code>commons_messaging</code> (L1)	partial	null:// adapter fully working; Telegram + Claude Code stubs in place (HRD-011/012 for live wiring). 9 SQL migrations embedded; pgx pool + RLS-enforcing WithTenantContext live (commit c8168ec); HRD-010 (commits 82ea82d..13cea69) wired queue (digital.vasic.background → pgxTaskRepository) + Redis (digital.vasic.cache) + the pherald migrate up/status/down/validate CLI. The §107 covenant proved its worth here: the E14 RLS falsifiability test caught (and the implementer fixed) a real production RLS-bypass bug where the bootstrap PG superuser silently bypassed FORCE ROW LEVEL SECURITY. HRD-085..090 opened to track the remaining 16 pgxTaskRepository methods returning ErrUnsupported.
<code>commons_storage</code> (L1)	✓ landed (M2 + HRD-010)	
<code>commons_infra</code>	✓ landed	QuickstartBoot + WithTenantContext + on-demand container orchestration via <code>containers/</code> .
<code>commons/cli</code> (L0)	✓ landed (Wave 2 / HRD-092)	Shared CLI scaffold: NewRootCmd + VersionCmd + ServeCmd (with Middleware hook) + StubCmd + healthz/readyz/metrics handlers. Consumed by pherald + 6 new flavor binaries. Catalogue-check no-match → vendored Herald-internal per §11.4.74.
<code>pherald</code> (CLI)	✓ partial (M3 + Wave 2 refactor)	Cobra root + version + serve (live HTTP on :24791, /v1/healthz, /v1/readyz, /v1/events, /metrics) now consumes cli.ServeCmd with RequestIDMiddleware hook (commits 1f81d69, 31562cf). send/doctor/migrate/subscriber/deadletter stubbed with HRD pointers.
<code>sherald</code> (CLI)	✓ live (Wave 3a / HRD-098)	System Herald binary serving :24793 via cli.ServeCmd + commons_auth.GinMiddleware (JWT gate). /v1/safety_state now LIVE — backed by sherald/internal/safety (process-local Aggregator + 15s mem-sampler

Module	Status	Notes
cherald (CLI)	 live (Wave 3a / HRD-028)	goroutine, lifecycle bound to SIGTERM-cancelled context). 5 §43 stubs (HRD-033/034/040/046/056) still return honest non-zero + HRD pointer. Constitution Herald binary serving :24792 via cli.ServeCmd + commons_auth.GinMiddleware (JWT gate). /v1/compliance now LIVE — backed by cherald/internal/compliance (paginated + filter-aware ConstitutionStore reads; tenant scope extracted from JWT claims). 11 §43 stubs still return honest non-zero + HRD pointer.
commons_auth	 landed (Wave 3a / HRD-099, renumbered from HRD-093)	JWT verifier — HMAC (HS256) + JWKS (RS256/ES256) hybrid; Redis-cached keys with in-memory fallback; Gin middleware + claims helpers. Catalogue-Check: no-match → vendored Herald-internal (docs/catalogue-checks/HRD-099-commons-auth.md).
bherald (CLI)	 scaffolded (Wave 2 / HRD-095)	Build Herald CLI-only binary. 3 §43 stubs (HRD-035/041/045).
rherald (CLI)	 scaffolded (Wave 2 / HRD-096)	Release Herald CLI-only binary. 3 §43 stubs (HRD-031/032/045).
iherald (CLI)	 scaffolded (Wave 2 / HRD-097)	Incident Herald binary serving :24794 via cli.ServeCmd. /v1/webhooks/page returns honest 501 + HRD-024.
scherald (CLI)	 scaffolded (Wave 2 / HRD-097)	Scheduled-audit Herald CLI-only binary. 1 §43 stub (HRD-047).

Operations

- **Repo hygiene:** main only; four-mirror fan-out proven across spec + Foundation M1/M2/M3 + anti-bluff infra + §107 covenant landing.
- **Constitution inheritance gate: 15 PASS / 0 FAIL** (was 12; I8a/b/c added 2026-05-20 for §107 covenant).
- **Anti-bluff audit: 16 PASS / 0 FAIL / 1 SKIP** (was 14; HRD-011 telebot.v3 third-party SKIP-with-reason per §11.4.74; I8 paired meta-test included).
- **CodeGraph validate:** 7 PASS / 0 FAIL / 2 SKIP (HRD-091 documented submodule-traversal gap).
- **E2E bluff-hunt against real services: 41 PASS / 0 FAIL / 5 SKIP** (was 33; Wave 3a added 8 new PASSes: E35/E36 auth gate × 2 flavors + E43/E44 cherald compliance live + E46/E47 sherard safety_state live; updated E28/E29 from "501 stub" → "401 JWT-gated" to reflect Wave 3a route swap). SKIPs are E17/E18/E34 (Telegram live, claude live, full inbound — requires operator credentials) + E45 (cross-binary cherald-reads-pherald-writes — Wave 3b once Runner lands) + E48 (sherard destructive-op trigger — HRD-033 body not implemented). Real Gin server, live /v1/* endpoints, real Postgres + Redis containers via containers/ submodule, M2 RLS integration tests + HRD-010 storage tests against the live stack, SIGTERM-graceful-shutdowns. Paired §1.1 mutation gates: tests/test_wave2_mutation_meta.sh (4/4 PASS — Wave 2 invariants) + tests/test_wave3_mutation_meta.sh (3/3 PASS — Wave 3a invariants M1+M5+post-flight, M6 SKIP). This is the canonical §107 evidence per Herald §107.

- **Submodules:** 12 vendored — 11 Helix-stack capability modules under `submodules/` (`auth`, `background`, `cache`, `config`, `database`, `eventbus`, `middleware`, `observability`, `recovery`, plus `Models + Concurrency` added during HRD-010) referenced via `replace` directives in consuming `go.mod` files, NOT via `go.work`; plus `containers/` (runtime auto-detection + on-demand container boot, consumed directly by Foundation tests + `pherald doctor`). All 12 carry the §11.4 anti-bluff anchor.
- **Build / test toolchain:** Go 1.25+ (verified on 1.26 per CLAUDE.md), Pandoc 3.9, WeasyPrint 66, Podman 5.8 — verified locally.

Risk surface

- **Submodule-catalogue discovery (operator mandate 2026-05-20).** Every new HRD MUST carry a Catalogue-Check: `reuse|extend|no-match <org/repo>@<sha>` line per Universal §11.4.74. New HRDs (HRD-029..056 §43 catalogue) currently default to TBD pending operator review.
- **Versioning discipline (operator mandate 2026-05-20).** Minor changes bump secondary version, major bump primary. Currently at V3 r7→r8 within the V3 series; a full V4 rewrite is reserved for a primary-version jump.
- **Live channel integrations untested at runtime.** HRD-011 (Telegram telebot live) requires operator-supplied credentials per `docs/CONTINUATION.md`. HRD-012 (Claude Code `claude --resume` live) is closed with live PASS evidence captured. The §107 covenant blocks claims of "done" on these until end-user evidence is captured.
- **HRD-008 operator-side quickstart not yet validated end-to-end on a fresh laptop.** The compose stack works in dev; operator hardware validation is the next concrete §107 evidence point.
- **No CI yet.** No GitHub Actions / GitLab CI / etc. Operator runs `scripts/e2e_bluff_hunt.sh` manually for now — that script IS the local CI surrogate.

Closed risks

- **Production RLS-bypass bug (closed 2026-05-20 during HRD-010 step 3, commit 5c1c022).** Until that commit, `commons_infra.WithTenantContext` set the tenant GUC but `ROW LEVEL SECURITY` was not FORCED on the targeted tables — and the bootstrap PG user used in tests was a superuser, which bypasses RLS regardless. The E14 falsifiability test (real Postgres, cross-tenant `SELECT` must return EXACTLY 0 rows from the other tenant) caught the bluff before any live deployment. Fix: switched the application connection to a non-superuser role + ran `ALTER TABLE ... FORCE ROW LEVEL SECURITY` in migration 000003. The §107 covenant directly prevented this from shipping; the lesson is "EXACT-N row count, never >=N" for any isolation test.

Recent activity

Wave 3a — substrate + 2 live REST routes (closed 2026-05-21)

8-step workstream commits `dbbea95` . . this shipped:

- `commons_auth/ JWT verifier scaffold` (HRD-099, renumbered from HRD-093; commit `dbbea95`) — new top-level Go module: HMAC (HS256) + JWKS (RS256/ES256) hybrid verifier with Redis-cached keys + in-memory fallback; Gin middleware (`GinMiddleware`) that gates every request behind the verifier and sets `ContextKeyClaims` on success; `claims.go` helpers (`TenantFromClaims`,

- SubjectFromClaims); verifier_test.go exercises both modes incl. expired-token + wrong-signature paths. Catalogue-Check: no-match → vendor Herald-internal (docs/catalogue-checks/HRD-099-commons-auth.md).
- **events_processed migration** (commit 3f8daed) — commons_storage/migrations/000012_events_processed.{up,down}.sql; RLS-enforced 30-day TTL inbound idempotency archive per V3 §32.2.
 - **ConstitutionStore.ListQuery extension** (commit 6276437) — Since/Until/Offset fields added; both Memory + Postgres backends honour them.
 - **cherald/internal/compliance/Handler + tests** (commit a2ef831) — paginated + filter-aware /v1/compliance handler; tenant scope from JWT claims; defaults page=1, page_size=50 (max 200).
 - **cherald wiring + route swap** (HRD-028 close-out, commit bfca740) — cherald/cmd/cherald/main.go builds the Verifier from env, instantiates the ConstitutionStore (Postgres if HERALD_PG_DSN else Memory), and wires commons_auth.GinMiddleware + compliance.Handler via cli.ServeOpts.Middleware. The 501-stub for /v1/compliance is gone.
 - **sherald/internal/safety/Aggregator + Handler + sampler** (commit 1016b96) — process-local Aggregator tracks lastMemPercent + lastDestructiveOp + uptime; StartMemSampler goroutine ticks every 15s; Handler returns the snapshot as JSON; tests cover the Aggregator surface.
 - **sherald wiring + sampler start** (HRD-098 close-out, commit bfa82bc) — sherald/cmd/sherald/main.go instantiates the Aggregator + starts the sampler under a SIGTERM-bound context; wires the Verifier + Handler via cli.ServeOpts.Middleware. The 501-stub for /v1/safety_state is gone.
 - **e2e + mutation gate + docs sync (commit 0cc6fad)** — 8 new e2e invariants (E35/E36/E43/E44/E46/E47 PASS; E45/E48 SKIP-with-reason); E28/E29 updated from "501 stub" to "401 JWT-gated"; e2e header set process-wide HERALD_AUTH_MODE + CODEGRAPH_ALLOW_UNSAFE_NODE; new paired §1.1 mutation gate tests/test_wave3_mutation_meta.sh (3/3 PASS); atomic Issues→Fixed for HRD-028, HRD-098, HRD-099 (commons_auth, renumbered from HRD-093 in this r11 cleanup); Status.md r9 → r10 → r11; Issues.md r8 → r9 → r10; Fixed.md r7 → r8 → r9.

Anti-bluff battery sign-off (this commit): inheritance gate 15/15 PASS; meta-tests (I6, I8, Wave2, Wave3) all green; audit_antibluff.sh 16 PASS / 0 FAIL / 1 SKIP (commons_auth not added to the audit's submodules/ walk — workspace-internal module by design); codegraph_validate.sh 7 PASS / 0 FAIL / 2 SKIP (with CODEGRAPH_ALLOW_UNSAFE_NODE=1 for Node 25 environment); e2e_bluff_hunt.sh 41 PASS / 0 FAIL / 5 SKIP.

Wave 2 — flavor-scaffold sweep (closed 2026-05-21)

15-step workstream commits 7e0a614..eef606b (with parallel logo branding 24b96f2) shipped:

- **Shared scaffold** — new commons/cli/ package: NewRootCmd (steps 2), VersionCmd (step 2), routes.go + healthz/readyz/metrics handlers (step 3), ServeCmd with Middleware-hook + graceful shutdown (step 4), StubCmd honest 501-style stubs (step 1). Vendored Herald-internal per §11.4.74 catalogue-check no-match → see docs/catalogue-checks/HRD-092-commons-cli.md.
- **pherald refactor** — main + stubs migrated to consume commons/cli/ (step 5); pherald serve consumes cli.ServeCmd with RequestIDMiddleware hook (step 6, commit 31562cf).
- **6 new flavor binaries** — sherald serving :24793 (step 7, HRD-093), cherald serving :24792 (step 8, HRD-094), iherald serving :24794 (step 9, HRD-097), bherald CLI-only

(step 10, HRD-095), rherald CLI-only (step 11, HRD-096), sgerald CLI-only (step 12, HRD-097). Each ships honest 501-style §43 command stubs with HRD pointers.

- **Anti-bluff additions** — `scripts/e2e_bluff_hunt.sh` grew from 18 → 33 invariants covering all 6 new binaries (version --json shape, healthz binding, /v1/safety_state + /v1/compliance + /v1/webhooks/page 501 honesty, sgerald SIGTERM graceful-shutdown, §43 stub exit-code, pherald regression sentinel). New paired mutation gate `tests/test_wave2_mutation_meta.sh` (4/4 PASS) proves the new invariants catch their claimed regressions.
- **Spec capture** — V3 r7 → r8 added §6.3 Branding extension (per-flavor fields), §18.0 Wave 2 catalogue, §41 REST route additions, §44.M Wave 2 milestone, §43.2 catalogue-check addendum (step 14, commit `ee f606b`).
- **Atomic Issues→Fixed** — HRD-092..097 closed in r7 of Fixed.md; HRD-098 opened in r8 of Issues.md as Wave 3+ deferred (sgerald /v1/safety_state live impl).

Anti-bluff battery sign-off (this commit): inheritance gate 15/15 PASS; meta-tests (I6, I8, Wave2) all green; `audit_antibluff.sh` 16 PASS / 0 FAIL / 1 SKIP; `codegraph_validate.sh` 7 PASS / 0 FAIL / 2 SKIP; `e2e_bluff_hunt.sh` 33 PASS / 0 FAIL / 3 SKIP.